

Inequalities – 5

Matrix and spectral inequalities

Homework

10 сентября 2026 года

Theoretical minimum. The Frobenius inner product and Cauchy–Bunyakovsky for the trace; von Neumann’s trace inequality; Hadamard’s inequality; concavity of $\log \det$ on positive definite matrices and the Minkowski determinant inequality; Weyl’s inequalities and the Hoffman–Wielandt bound; the variational description of the eigenvalues.

1. Prove that $\|AB\|_F \leq \|A\| \cdot \|B\|_F$ for every pair of matrices, and that the singular values satisfy $\sigma_{i+j-1}(AB) \leq \sigma_i(A)\sigma_j(B)$; deduce $\|A^k\| \leq \|A\|^k$ and give an example where the inequality is strict.
2. (Proposed problems) Let $n \geq 2$. (a) Prove that $\det((A^2 - B^2)(C^2 - B^2)) \geq 0$ for all $A, B, C \in M_n(\mathbb{R})$ with $AB = BC$. (b) Find all $k \geq 1$ for which $\det((A^k - B^2)(C^k - B^2)) \geq 0$ holds for all such A, B, C .
3. (IMC) For an $n \times n$ real matrix M let $\|M\|$ be the operator norm with respect to the Euclidean norm. Assume $\|A^k - A^{k-1}\| \leq \frac{1}{2002k}$ for all positive integers k . Prove that $\|A^k\| \leq 2002$ for all k .
4. (IMC 2017) Let A be an $n \times n$ complex matrix all of whose eigenvalues have absolute value at most 1. Prove that $\|A^n\| \leq \frac{n}{\ln 2} \|A\|^{n-1}$.
5. Prove Fischer’s inequality: for a positive definite block matrix $\begin{pmatrix} A & B \\ B^* & C \end{pmatrix}$ one has $\det \leq \det A \cdot \det C$, and deduce Hadamard’s inequality by induction.
6. Prove Kantorovich’s inequality: for a positive definite A with eigenvalues in $[m, M]$ and any unit vector x , $\langle Ax, x \rangle \langle A^{-1}x, x \rangle \leq \frac{(m+M)^2}{4mM}$. Deduce that $\operatorname{tr} A \cdot \operatorname{tr} A^{-1} \geq n^2$, with equality exactly for scalar matrices.

7. (OMOUS 2023) Let $a_1, \dots, a_n, b_1, \dots, b_n$ be unit vectors in $\mathbb{R}^n$. Show that there are signs $\varepsilon_i = \pm 1$ such that the matrix C with columns $c_i = a_i + \varepsilon_i b_i$ satisfies $|\det C| \leq 2^{n/2}$.
8. Prove the Hoffman–Wielandt inequality for symmetric matrices, and deduce that the map sending a symmetric matrix to its ordered spectrum is 1-Lipschitz from the Frobenius norm to the Euclidean norm.
9. Prove that $\operatorname{tr}(AB) \leq \frac{1}{2} \operatorname{tr}(A^2 + B^2)$ for symmetric A, B , that $\operatorname{tr}(AB) \geq 0$ when both are positive semidefinite, and give an example of positive semidefinite A, B with AB not symmetric.
10. Prove the Cauchy interlacing theorem for a principal submatrix of a symmetric matrix, and deduce that the largest eigenvalue is at least the largest diagonal entry and at most the maximal row sum of absolute values.